# Personal Psychology 20

Unit 6

nextSTEP
SENIOR HIGH SCHOOL
Fort Saskatchewan 780 992 0101
Sherwood Park 780 464 1899
Vegreville 780 632 7998

STUDENT NAME:

Value
24

## Part One: Multiple-Choice Items

Read each item carefully and decide which choice best completes the statement or answers the question. Write the letter of your choice in the blank before the item.

_____ 1. The human brain is separated into _____ hemispheres with _____ lobes each.

A. two; two
B. two; four
C. four; two
D. four; four

_____ 2. The hemispheres of the brain are connected by a

A. synapse
B. connecting lobe
C. corpus callosum
D. permeable bundle

_____ 3. Phineas Gage's brain injury cause changes in his

A. hearing
B. personality
C. language skills
D. ability to sense cold

_____ 4. The maintenance of life-sustaining body processes beyond our conscious control is specifically controlled by the _____ nervous system.

A. central
B. automatic
C. distributed
D. autonomic

_____ 5. The _____ division of the _____ nervous system deals with tasks such as waste elimination and the conservation of energy.

A. sympathetic; central
B. sympathetic; autonomic
C. parasympathetic; central
D. parasympathetic; autonomic

_____ 6. Neurons release special chemicals called _____ in the spaces between cells; the spaces are called _____.

A. neurotransmitters; synapses
B. electrotransmitters; synapses
C. neurotransmitters; nodes of Ranvier
D. electrotransmitters; nodes of Ranvier

_____ 7. Epilepsy is caused by

A. convulsions
B. degeneration of myelin
C. degeneration of the Substantia Nigra
D. abnormal electrical impulses in the brain

_____ 8. Alzheimer's disease is characterized by a loss of _____ and usually occurs in people between _____ years of age.

A. memory; 40-50
B. memory; 60-70
C. motor control: 40-50
D. motor control; 60-70

_____ 9. Scientists are aware of the functions different lobes of the brain perform because they have studied

A. MRI brain scan results
B. autopsied brains of criminals
C. the effects of brain injuries in living people
D. the start points of neurons in the brain to the end points in specific organs

_____ 10. Electroencephalograms measure

A. magnetic pulses
B. radioactive glucose
C. brain wave activity
D. the mass of brain tumours

_____ 11. The sudden and unpredictable onset of sleep is called

A. insomnia
B. narcolepsy
C. sleep apnea
D. pseudoinsomnia

_____ 12. The inability to fall asleep is called

A. insomnia
B. narcolepsy
C. sleep apnea
D. pseudoinsomnia

_____ 13. There are _____ distinct sleep stages.

A. three
B. four
C. five
D. six

_____ 14. Which of the following sleep disorders is linked to REM sleep?

A. Insomnia
B. Narcolepsy
C. Night terrors
D. Sleep walking

_____ 15. Which of the following is **not** an effect of sleep deprivation?

A. Severe night terrors
B. Decreased immunity
C. Impaired mental ability
D. An increasing sleep debt

_____ 16. Which of the following wave types are not detected in our brains when we sleep?

A. beta
B. delta
C. theta
D. gamma

_____ 17. In stages _____ of sleep, a person may sleep walk and experience bed-wetting.

A. one and two
B. two and three
C. three and four
D. four and five (REM)

_____ 18. In stage _____ of sleep, a person may experience the feeling of falling and jerk him/herself awake.

A. one
B. two
C. three
D. four

_____ 19. In this stage of sleep, dreaming occurs.

A. Two
B. Three
C. Four
D. Five (REM)

_____ 20. When we dream, our heart rate _____ and our arm & leg muscles become _____.

A. increases; active
B. decreases; active
C. increases; paralyzed
D. decreases; paralyzed

_____ 21. This group spends the most time in stage five or REM sleep.

A. Infants/babies
B. Teenagers
C. Adults
D. Elderly

_____ 22. Species-specific behaviours are _____ behaviours much like _____.

A. learned; habits
B. learned; instincts
C. unlearned; habits
D. unlearned: instincts

_____ 23. This is the phenomenon of baby birds following the first thing they see that moves.

A. Imitation
B. Releasing
C. Imprinting
D. Emancipating

_____ 24. This is a term for the unlearned behaviour patterns that appear in the same form at a certain point of development of every member of a species.

A. Reflex
B. Instinct
C. Automation
D. Predisposition

**Value 10**

## Part Two: Matching

Choose the letter of the term that best describes the concept or completes the sentence. Terms may not be used more than once and not all terms will be used.

**Statements**

_____ 1. This lobe is responsible for the body's ability to sense touch, pressure, and temperature.

_____ 2. This substance has been linked to aggression.

_____ 3. This chemical helps manage emotional arousal, learning, and memory.

_____ 4. This lobe is responsible for hearing.

_____ 5. A lack of this substance is thought to be responsible for causing depression in some individuals.

_____ 6. This chemical increases heart rate and blood pressure.

_____ 7. These have specific shapes so they can fit into a receptor site specifically designed for it; like a specific key fits a specific lock.

_____ 8. The _____ division of the autonomic nervous system is responsible for responding to crisis situations.

_____ 9. People with this disorder have difficulty with maintaining body posture and producing speech as well as display involuntary body movements.

____ 10. This hormone increases in females after giving birth.

**Terms**

A. acetylcholine
B. Alzheimer's disease
C. dopamine
D. electrotransmitters
E. epilepsy
F. epinephrine
G. estradiol
H. Frontal
I. neurotransmitters
J. multiple sclerosis
K. parasympathetic
L. Parietal
M. Parkinson's disease
N. progesterone
O. serotonin
P. sympathetic
Q. temporal
R. testosterone

Value 10

## Part Three: True or False

Read each item carefully and decide if the statement is true or false. Write **T** or **F** in the blanks to indicate your choices.

_____ 1. A severe injury to the occipital lobe may result in blindness.

_____ 2. Lobotomies are still often used to control behaviour problems in mental patients.

_____ 3. The left hemisphere controls the right side of the body.

_____ 4. The time between the brain sending a signal to a muscle and that muscle contracting is instantaneous.

_____ 5. Positron Emission Tomography scans or PET scans use a radioactive solution that binds to glucose in the blood stream.

_____ 6. The endocrine system consists of glands that release hormones that control body temperature, gender of a fetus, and the moods of both men and women.

_____ 7. The hormone estrogen is especially important in regulating nest building instincts in some animals.

_____ 8. Special brain cells called neurons do not directly touch each other.

_____ 9. High fevers can bring on epileptic attacks.

___ 10. An MRI scan can show whether cells have enough oxygen and water.

Value 16

## Part Four: Written-Response Items

Respond to the following items on the blanks below the items.

**2** 1. How does cataplexy differ from narcolepsy?

_____

_____

_____

_____

_____

2. Using the sleep survey that follows, question/survey at least four individuals over the age of 12 (e.g., parents, siblings, other relatives, friends, neighbors, etc.), then respond to the items that follow. Make sure to record their responses in the following table.

## Sleep Survey

| | |
|---|---|
| **Q1.** | Snoring occurs when there is an obstruction to the free flow of air through the passages at the back of the mouth and nose. |
| | A: TRUE – This area is the collapsible part of the airway where the tongue and upper throat meet the soft palate and uvula. Snoring occurs when these structures strike each other and vibrate during breathing. |
| **Q2.** | Ten percent of adults snore regularly. |
| | A: FALSE – Current studies indicate that approximately 25% of adults sore regularly. |
| **Q3.** | Snoring is often an indication of a medical problem. |
| | A: TRUE – When snoring is severe, it can be associated with serious, long-term health problems, including obstructive sleep apnea. |
| **Q4.** | Obstructive sleep apnea occurs when breathing stops completely due to an obstructed airway. |
| | A: TRUE – When loud snoring is interrupted by frequent episodes of totally obstructed breathing, it is known as obstructive sleep apnea. |
| **Q5.** | Having an alcoholic beverage before bed can help people relax so they can have a good night's sleep. |
| | A: FALSE – Alcohol may help a person enter light sleep, but it prevents a person from entering REM sleep |
| **Q6.** | Beverages containing caffeine, such as coffee, can cause insomnia. |
| | A: TRUE – Caffeine is a stimulant. |
| **Q7.** | Teens need just as much sleep as babies do. |
| | A: FALSE – On average, babies need 16 hours of sleep per day while teens need 9 hours. |
| **Q8.** | When tested on a driving simulator, people who are sleep deprived perform just as poorly as individuals who are intoxicated. |
| | A: TRUE – And the worst combination is someone who is both sleep deprived and intoxicated! |
| **Q9.** | Narcolepsy is a sudden sleep attack in the middle of being awake. |
| | A: TRUE – Individuals cannot control the onset of a sleep attack. |
| **Q10.** | Narcolepsy is caused by not getting enough sleep. |
| | A: FALSE – A person with narcolepsy can still experience sleep attacks despite receiving sufficient sleep the night before. |

**6** a. Complete the table below summarizing the responses you obtained from each participant for each question. Include the total score of each individual in the final row.

| **Question Number (and correct answer)** | **Person One** | **Person Two** | **Person Three** | **Person Four** |
|---|---|---|---|---|
| Example Question *(F)* | ✓ | x | X | ✓ |
| Question 1 *(T)* | | | | |
| Question 2 *(F)* | | | | |
| Question 3 *(T)* | | | | |
| Question 4 *(T)* | | | | |
| Question 5 *(F)* | | | | |
| Question 6 *(T)* | | | | |
| Question 7 *(F)* | | | | |
| Question 8 *(T)* | | | | |
| Question 9 *(T)* | | | | |
| Question 10 *(F)* | | | | |
| **Total Score** (Number Correct out of 10) | | | | |

**3** b. In which areas were your participants **least** knowledgeable? For example, where there any questions where fewer than two individuals answered the question correctly?

**3** c. In which areas were your participants **most** knowledgeable? For example, where there any questions where more than two individuals answered the question correctly?

**2** d. Provide two reasons why you think sleep is an important activity.

_____

_____

_____

_____

_____

_____

_____

**End of Assignment Booklet 6**
**After you have completed all items, submit your work to your marker.**